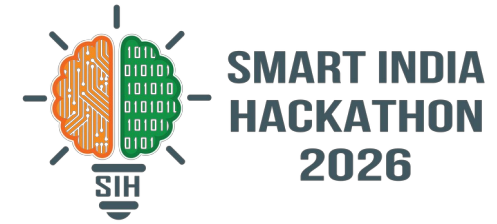


SMART INDIA HACKATHON 2026



RAILSYNC

AI-Powered Railway Maintenance Block Planning

SMART INDIA HACKATHON 2026 - IDEA SUBMISSION

Problem Statement ID: SIH26027

Problem Statement Title: AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways

Theme: Transportation & Logistics

PS Category: Software

Team ID: To be allotted after internal hackathon

Team Name: RailSync



Constraint-aware planning | Integrated possessions |
Recovery

PROPOSED SOLUTION

Detailed Solution

- Combines timetable-derived train occupancy with Engineering, S&T and TRD maintenance demand.
- Checks crew, machine, power, deadline and section-capacity constraints before granting a block.
- Uses CP-SAT to generate feasible coordinated maintenance possessions.
- Supports monthly, weekly, day-of planning and disruption recovery.

How It Addresses the Problem

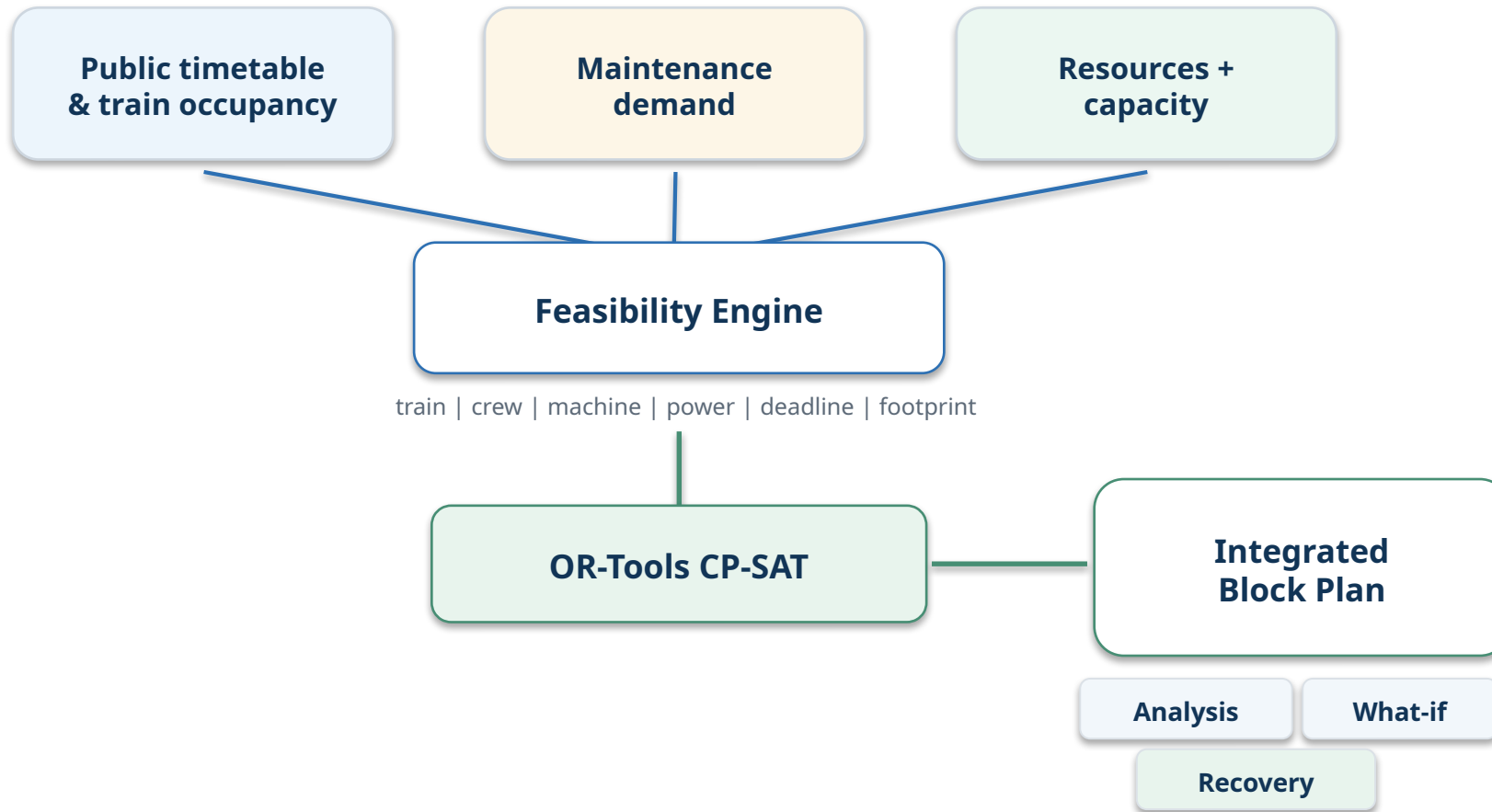
- Rejects gaps that are traffic-free but operationally infeasible.
- Prioritizes critical, urgent and overdue maintenance within hard constraints.
- Coordinates compatible departmental work into shared possessions.
- Re-optimizes affected work when trains or resources change.

Innovation & Uniqueness

- Multi-department integrated possessions reduce repeated infrastructure closure.
- Capacity-aware, multi-section planning without assuming direction equals track.
- Solver-backed alternatives, what-if analysis and explainable decisions.
- AI Copilot explains the plan while CP-SAT remains the scheduling authority.

OUTCOME: Complete priority maintenance safely with fewer, better-coordinated possessions.

SYSTEM FLOW



WORKING PROTOTYPE

3 public timetable territories

- Eastern: Saktigarh - Memari
- Western: Virar - Dahanu Road
- Delhi/Northern: Hazrat Nizamuddin - Palwal
- Multi-section & capacity footprints
- Plan + block lifecycle controls
- Alternatives, what-if & multi-disruption recovery
- Import validation and CSV/print export

TECHNOLOGY STACK

React + Vite

FastAPI

Python

OR-Tools

WORKING EVIDENCE, RISKS AND MITIGATION

Feasibility

- Working end-to-end prototype across three public corridor demonstrations.
- CP-SAT handles train, deadline, crew, machine, power and capacity constraints.
- Multi-section possessions, lifecycle, alternatives, what-if and recovery are implemented.
- Architecture is adapter-based so data feeds can be replaced without rewriting the solver.

Potential Challenges

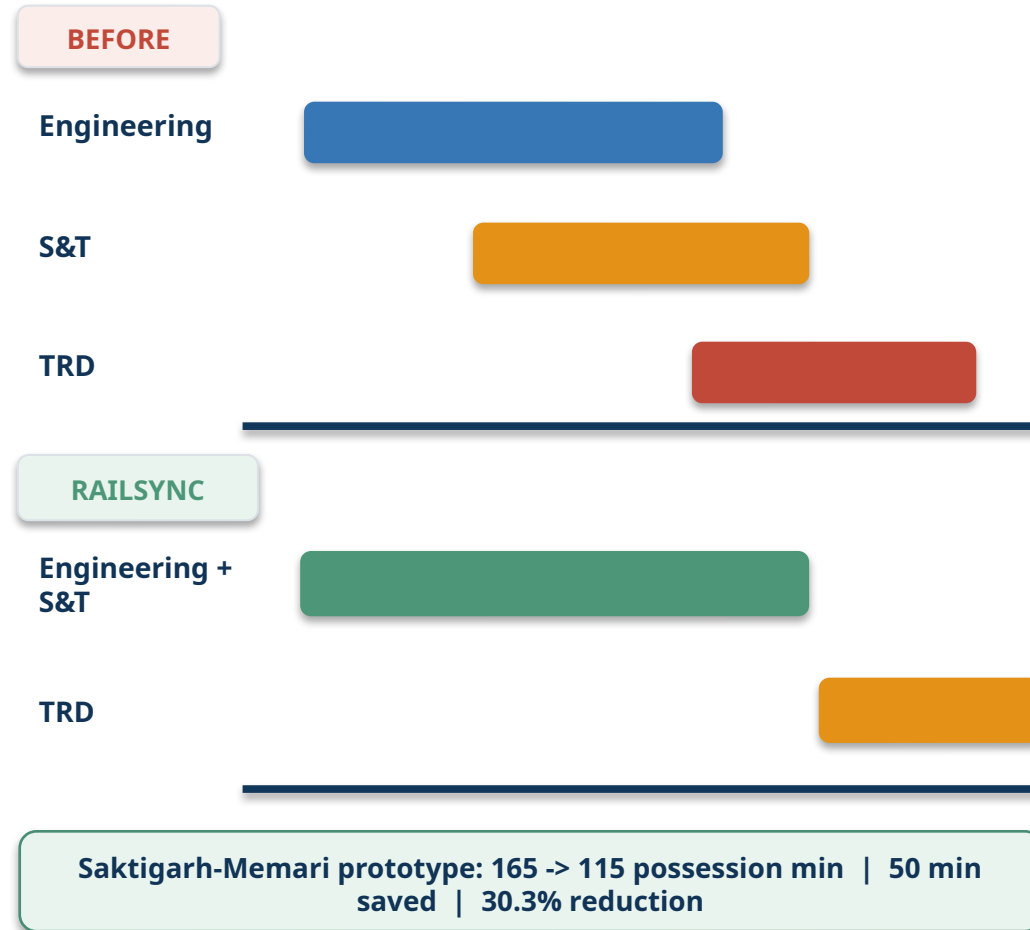
- Internal live maintenance and resource feeds are not publicly available.
- Detailed track topology and operational rules vary by territory.
- Day-of train or resource disruptions can invalidate an approved plan.

Mitigation Strategy

- Canonical adapters are ready for authorized Railway feeds when available.
- Conservative whole-section protection is used when infrastructure detail is unknown.
- Solver-backed recovery preserves immutable work and changes only affected possessions.
- Data provenance and assumptions are documented for every demonstration territory.

VERIFICATION: 218 Python tests passed | 8 frontend tests passed | data validation passed | lint + production build passed | browser flow verified

FROM FRAGMENTED BLOCKS TO COORDINATED POSSESSIONS



USER STORY

Engineering and S&T both need overnight work on the same corridor while trains continue to operate.

RailSync finds a common feasible window, checks resources and deadlines, and combines compatible work into one possession. If a train is delayed, only affected work is re-planned.

IMPACT

- Higher infrastructure and asset availability
- Lower total possession time for the same maintenance
- Better Engineering / S&T / TRD coordination
- Faster, minimum-change disruption recovery
- Explainable and auditable planning decisions

OFFICIAL DATA SOURCES, OPTIMIZATION AND REPOSITORY

Eastern Railway

[Combined Suburban Time Table](#)[Eastern Railway - historical public timetable](#)

Western Railway

[Western Railway timetable + Dahanu Road UP timetable](#)westernrailway.indianrailways.gov.in

Northern Railway

[Northern Railway System Map 2025 + timetable references](#)nr.indianrailways.gov.in

Optimization Engine

[Google OR-Tools - CP-SAT Solver](#)developers.google.com/optimization

PROJECT REPOSITORY

github.com/prakhar9642/RailSync

Working prototype | source code | documentation | provenance



SCAN REPO